

1.1.

$$1. \operatorname{Re}(\langle u, v \rangle) = \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2)$$

$$\begin{aligned} \|u+v\|^2 &= \langle u+v, u+v \rangle = \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \\ &+ \langle v, v \rangle = \|u\|^2 + \|v\|^2 + \langle u, v \rangle + \langle v, u \rangle \end{aligned}$$

$$\|u-v\|^2 = \langle u-v, u-v \rangle = \langle u, u \rangle - \langle u, v \rangle + \langle -v, u \rangle +$$

$$\begin{aligned} &+ \langle -v, -v \rangle = \langle u, u \rangle - \langle u, v \rangle - \langle v, u \rangle + \langle v, v \rangle \\ &= \|u\|^2 + \|v\|^2 - \langle u, v \rangle - \langle v, u \rangle \end{aligned}$$

$$\|u+v\|^2 - \|u-v\|^2 = 2 (\langle u, v \rangle + \langle v, u \rangle) \quad (1.)$$

obecně platí $2 \operatorname{Re}(x) = x + \bar{x}$

$$2 \operatorname{Re}(\langle u, v \rangle) = \langle u, v \rangle + \overline{\langle u, v \rangle} = \langle u, v \rangle + \langle v, u \rangle$$

$$\Rightarrow \operatorname{Re}(\langle u, v \rangle) = \frac{1}{2} (\langle u, v \rangle + \langle v, u \rangle),$$

$$\text{ale } \frac{1}{4} (\|u+v\|^2 - \|u-v\|^2) = \frac{1}{2} (\langle u, v \rangle + \langle v, u \rangle)$$

2 (1.)

□

2.

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2)$$

$$\|u - iv\|^2 = \langle u - iv, u - iv \rangle =$$

$$= \langle u, u \rangle - i \langle u, v \rangle + i \langle v, u \rangle + \langle u, v \rangle$$

$$\|u + iv\|^2 = \langle u + iv, u + iv \rangle =$$

$$= \langle u, u \rangle + i \langle u, v \rangle - i \langle v, u \rangle + \langle u, v \rangle$$

$$\Rightarrow \|u - iv\|^2 - \|u + iv\|^2 = -2i \langle u, v \rangle + 2i \langle v, u \rangle \quad (2.)$$

avec une autre place' $2i \operatorname{Im}(x) = x - \bar{x}$, a l'eq

$$\operatorname{Im}(\langle u, v \rangle) = \frac{1}{2i} (\langle u, v \rangle - \overline{\langle u, v \rangle}) =$$

$$= \frac{1}{2} (-i \langle u, v \rangle + i \langle v, u \rangle), \text{ avec une autre,}$$

$$\text{à l'eq } \frac{1}{4} (\|u - iv\|^2 - \|u + iv\|^2) =$$

$$= \frac{1}{2} (-i \langle u, v \rangle + i \langle v, u \rangle) \quad \text{à (2.)}$$

□